DO YEONG KANG

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RESEARCH INTEREST

- **Efficient AI & Model Compression**
- **Multi Agent System (MAS) & Multi Agent Communication**
- **Matrix & Tensor Decomposition**
- **Hyperdimensional Computing (HDC) & In-Memory Computing (IMC)**

EXPERIENCE

Present | 2026 Sep.
Ph.D. in Electrical and Computer Engineering
Georgia Institute of Technology, Atlanta, GA, USA

- Multi agent latent communication
- Token-adaptive dynamic SVD [*C4*]
- Efficient MoE training & MoE architecture based machine unlearning [*C3*]
- Dimension pruning for compact HDC-IMC implementations [*J1*] [*A4*]

Advised by Prof. Saibal Mukhopadhyay

2026 Aug. | 2024 Sep.
M.S. in Electrical and Computer Engineering
Sungkyunkwan University (SKKU), Suwon, Korea

- Token-adaptive dynamic SVD [*C4*]
- Efficient MoE training & MoE architecture based machine unlearning [*C3*]
- Dimension pruning for compact HDC-IMC implementations [*J1*] [*A4*]

Advised by Prof. Jong Hwan Ko & Dr. Kang Eun Jeon

2024 | 2023
Research Intern
Sungkyunkwan University (SKKU), Suwon, Korea

- Researched emerging brain-inspired paradigms HDC & Vector Symbolic Architectures (VSA)
- Designed IMC array-optimized HDC algorithmic structure with full array utilization [*C2*]
- Optimized BatchNorm for Once-for-All Networks, achieving training time reduction [*A2*]

Advised by Prof. Jong Hwan Ko & Dr. Kang Eun Jeon

2022 | 2020
KATUSA (Korean Augmentation To the United States Army)
Pyeongtaek, Korea

- Provided interpretation and translation support between U.S. and ROK Army

PUBLICATIONS

[1]: Equal contribution, *: Corresponding author

C5 **D. Y. Kang**, S. H. Han, J. G. Choo, K. E. Jeon*, and J. H. Ko*, "Dynamic SVD via Token-Adaptive Basis Selection," *Neural Information Processing Systems (NeurIPS)*, 2026. [under review]

C4 K. E. Jeon, Y. S. Kang, **D. Y. Kang**, T. Y. Lee, G. M. Park*, and J. H. Ko*, "R-ESC: Robustly Erasing Space Concepts via Stochastic Feature Remapping," *European Conference on Computer Vision (ECCV)*, 2026. [will be appeared]

C3 Y. H. Oh, **D. Y. Kang**, J. Park, C. Hwang, K. E. Jeon*, and J. H. Ko*, "HyperSPACE: Sparse-Adder-Compatible Encoding for Efficient Hyperdimensional Computing on Digital CIM Arrays," *IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2026. [will be appeared]

C2 **D. Y. Kang**, Y. H. Oh, C. Hwang, J. Kim, K. E. Jeon*, and J. H. Ko*, "MEMHD: Memory-Efficient Multi-Centroid Hyperdimensional Computing for Fully-Utilized In-Memory Computing Architectures," *Design, Automation and Test in Europe Conference (DATE)*, 2025. [Paper]

C1 E. J. Hwang, **D. Y. Kang**, J. H. Moon, H. Y. Seong, S. W. Lee, and J. W. Jeon*, "Implementation of Autonomous Parking System Using LiDAR-based Triangulation Method," *Korea Information Processing Society*, 2023. [Paper]

J1 **D. Y. Kang**[1], Y. H. Oh[1], C. Hwang, J. Kim, K. E. Jeon*, and J. H. Ko*, "Multi-Centroid Hyperdimensional Computing for Compact IMC Arrays via Dimension Pruning," *IEEE Internet of Things Journal*, [will be appeared]

HONORS & AWARDS

A6 **On-Device AI Optimization Challenge –** *Nota AI (2026)*
Awarded 2nd place (out of 24 teams) at NetsPresso On-Device LLM Runtime Optimization track [Post]

A5 **Samsung Outstanding Poster Award –** *Samsung Electronics (2025)*
Top voted project for Samsung Electronics Industry-Academia Collaboration Exchange

A4 **DAC Young Fellows –** *Design Automation Conference (2025)*
Selected for emerging researchers in Electronic Design Automation (EDA) field

A3 **BK Scholarship –** *BrainKorea21 (2024-2025)*
Granted $10k per year for outstanding graduate students in Department of Electrical and Computer Engineering

A2 **Cross-disciplinary Research Award –** *Sungkyunkwan University (2024)*
Awarded 2nd place (out of 16 teams) for cross-disciplinary research with students from diverse majors.

A1 **SKKU Autonomous Driving SW Competition –** *Hyundai Mobis & Sungkyunkwan University (2023)*
Awarded 3rd place (out of 25 teams) in autonomous driving and parking challenges hosted by Hyundai Mobis.

PATENTS

P2 **D. Y. Kang**, S. H. Han, J. H. Ko and Samsung Electronics, KR Patent Application (TBD), SVD-based Compression of Large Language Models via Token-Adaptive Dynamic Basis Selection, Jul. 2026.

P1 **D. Y. Kang**, Y. H. Oh, C. Hwang, K. E. Jeon, J. H. Ko and H. H. Lee, KR Patent Application (10-2024-0174593), Memory-efficient hyperdimensional computing techniques for in-memory computing optimization, Nov. 2024.

TEACHING EXPERIENCE

2026 **Data Structures and Algorithms** *(teaching assistant)*
SungKyunKwan University (SKKU), Suwon, Korea

- Served as a teaching assistant (TA) for an undergraduate course with 50+ students

2026, 2025 **On-device AI** *(teaching assistant)*
SungKyunKwan University (SKKU), Suwon, Korea

- Delivered lectures to 100+ undergraduate and graduate students in the Next-Generation Vehicle Bootcamp
- Developed and delivered lectures to explain core principles of efficient machine learning
- Conducted hands-on sessions on GPTQ and SparseGPT [Lab1] [Lab2]

SKILLS

- **Programming Languages:** Python, C, C++, MATLAB
- **Tools:** PyTorch, CUDA Programming, Triton, Huggingface, Numpy, TorchHD

REFERENCES

Saibal Mukhopadhyay *(Ph.D. advisor)*

Professor
School of Electrical and Computer Engineering
Georgia Institute of Technology, Atlanta, GA, USA

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Jong Hwan Ko *(M.S. advisor)*

Associate Professor
Department of Electrical and Computer Engineering
Sungkyunkwan University, Suwon, Korea

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Kang Eun Jeon

Post-doctoral Fellow
Kim Jaechul Graduate School of AI
Korea Advanced Institute of Science and Technology, Seoul, Korea

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kejeon@g.skku.edu